Lab 8 – Node js & React App

Lab Outcomes:

- Understand the purpose and significance of Node.js and React in modern web development.
- Successfully install Node.js and configure the environment.
- Create and run a basic React application.
- Gain familiarity with the structure of a React project.
- Develop foundational skills for building dynamic web applications using React.

Introduction:

In this lab, we will explore the fundamental concepts of **Node.js** and **React**, focusing on their installation processes and initial setup. Node.js, a JavaScript runtime built on Chrome's V8 JavaScript engine, allows developers to run JavaScript code outside the browser, making it a powerful tool for server-side development. React, on the other hand, is a widely-used JavaScript library for building dynamic and interactive user interfaces.

1. Node.js

Node.js is an open-source, cross-platform JavaScript runtime environment built on Chrome's V8 JavaScript engine. It enables developers to run JavaScript code outside the browser, making it suitable for server-side development. Node.js is widely used for building scalable and efficient server-side applications.We use Node.js for following reasons

- Allows JavaScript to run on the server.
- Non-blocking, event-driven architecture for high performance.
- Extensive package ecosystem via npm (Node Package Manager).
- Supports real-time applications like chats and streaming services.

2. React

- React is an open-source JavaScript library developed by Facebook for building user interfaces, specifically single-page applications (SPAs). It enables developers to create reusable UI components, making development faster and more efficient. We use react for
- Component-based architecture.
- Virtual DOM for faster rendering.
- Efficient state management.

- Large community and strong ecosystem.

3. What are npm and npx?

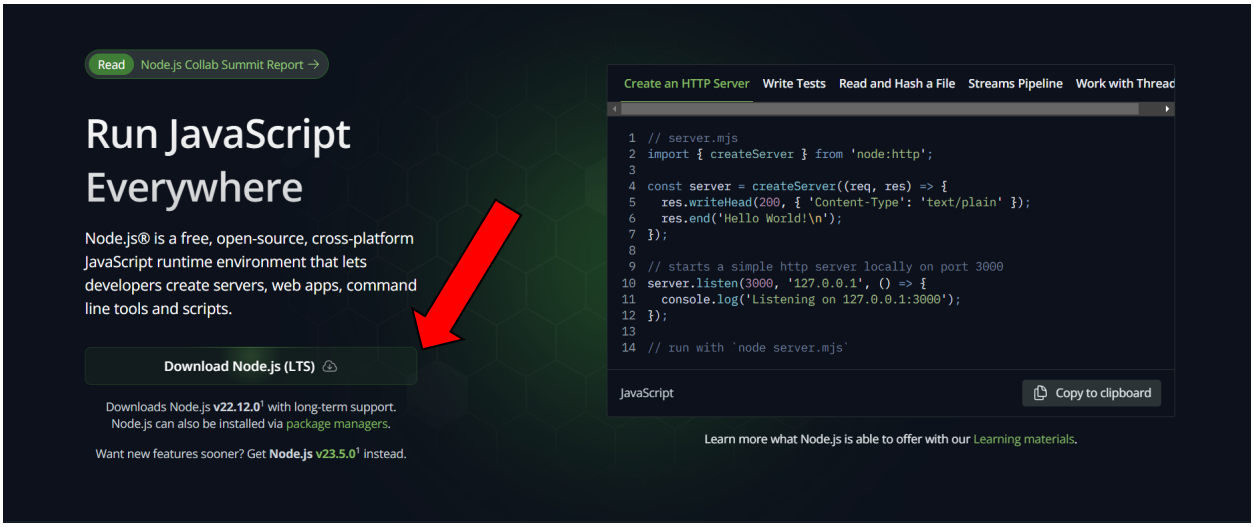
- **npm (Node Package Manager):** A package manager for Node.js, used to install and manage libraries, dependencies, and tools required for JavaScript projects.
- **npx (Node Package Runner):** A tool that comes with npm to execute Node packages directly without installing them globally.

Difference Between npm and npx:

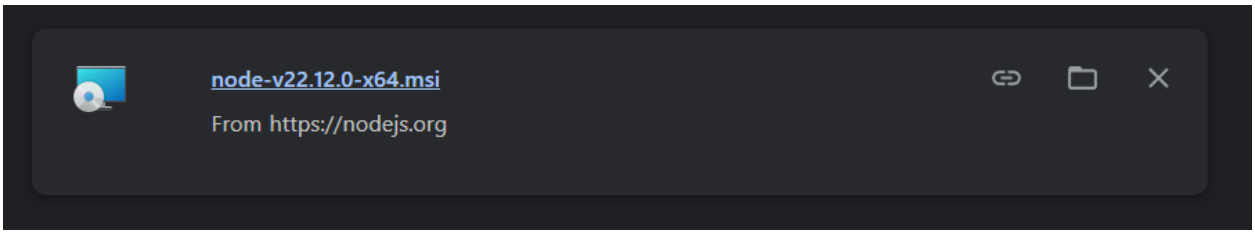
- npm installs and manages packages.
- npx directly runs packages without global installation.

Installation Tutorial:

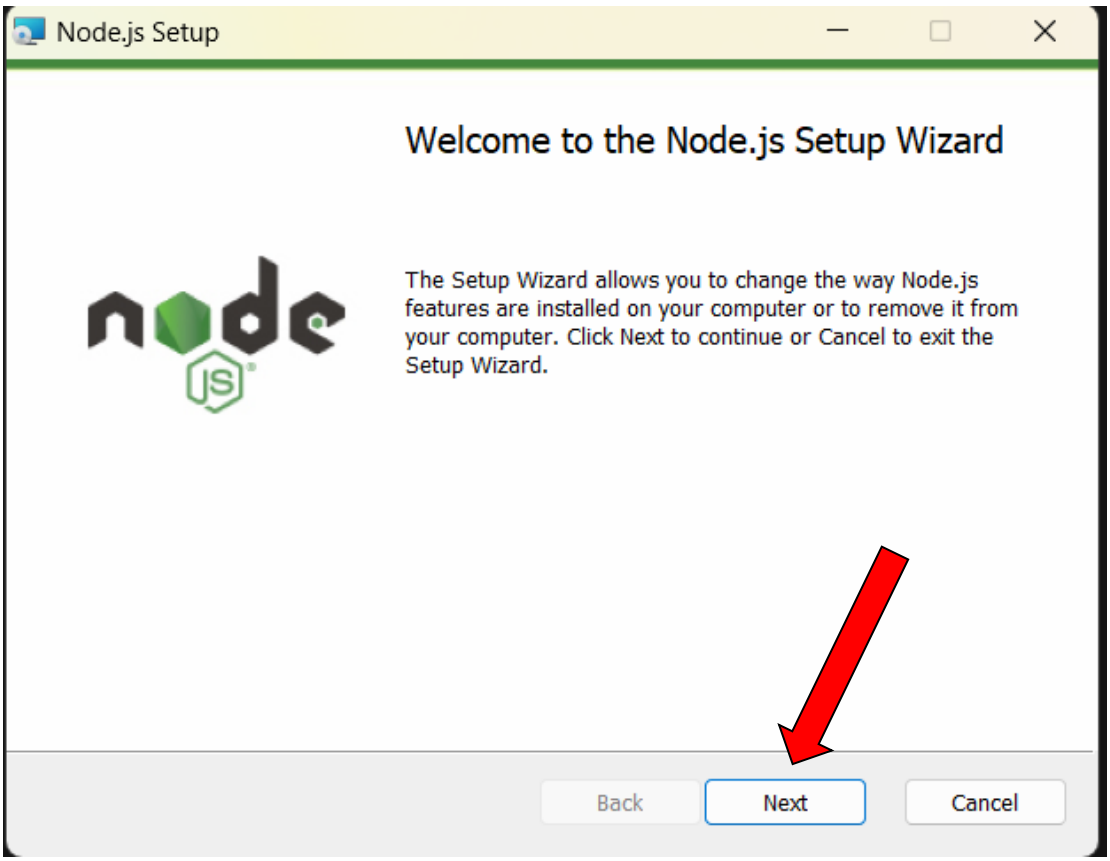
- Visit the official Node.js website: https://nodejs.org
- Download and install the LTS version by following the step by step guide given below.



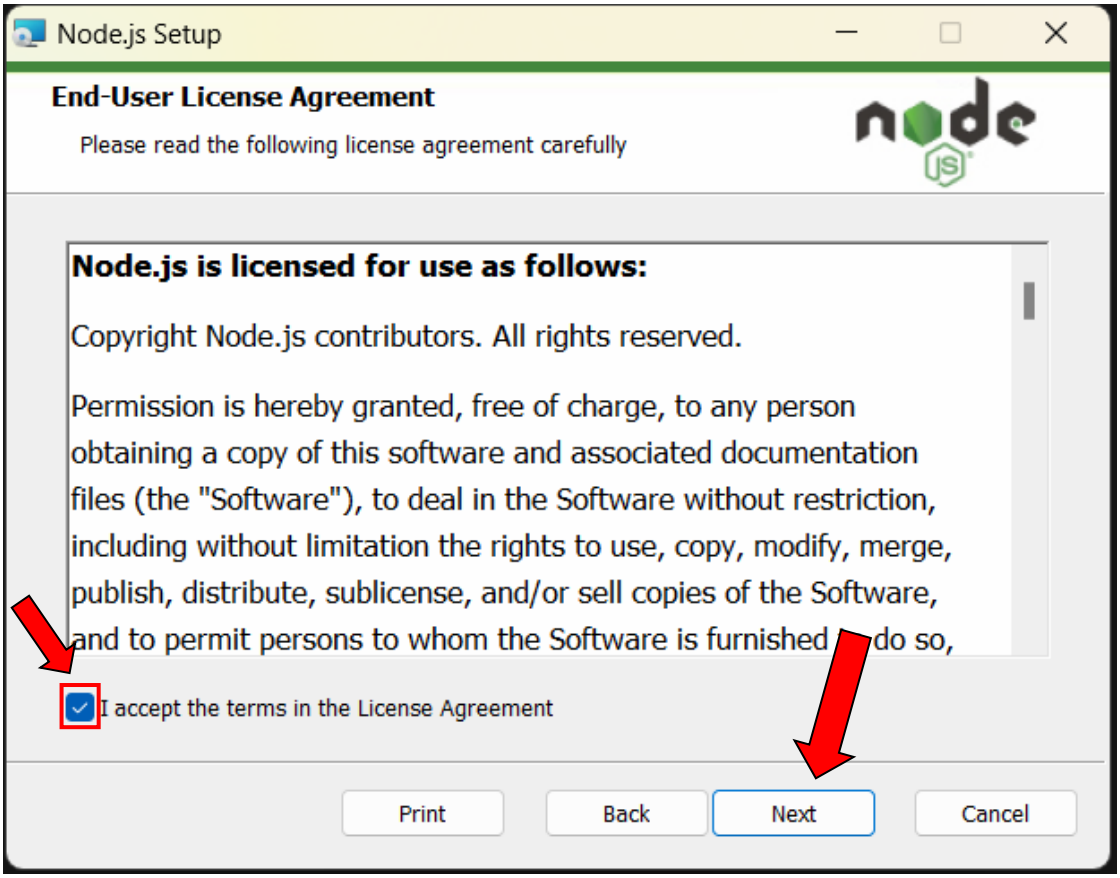
- Now Click on downloaded file



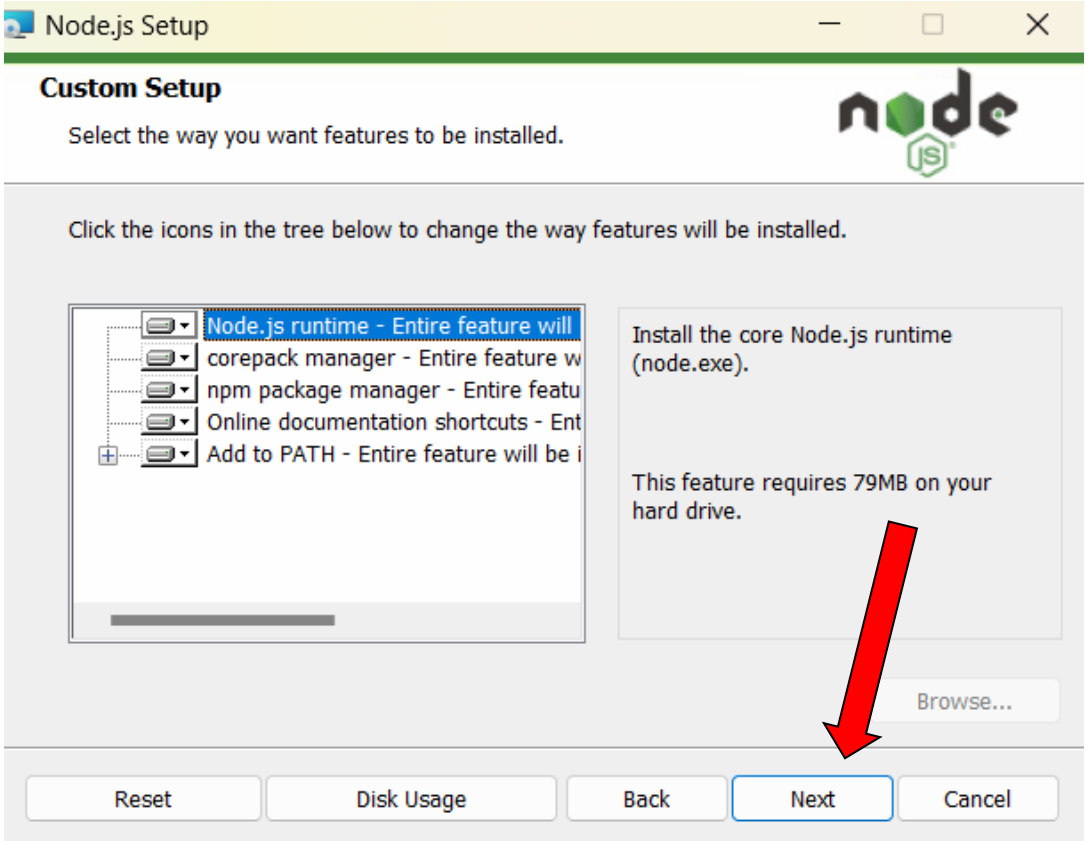
- After Clicking follow the installation instructions

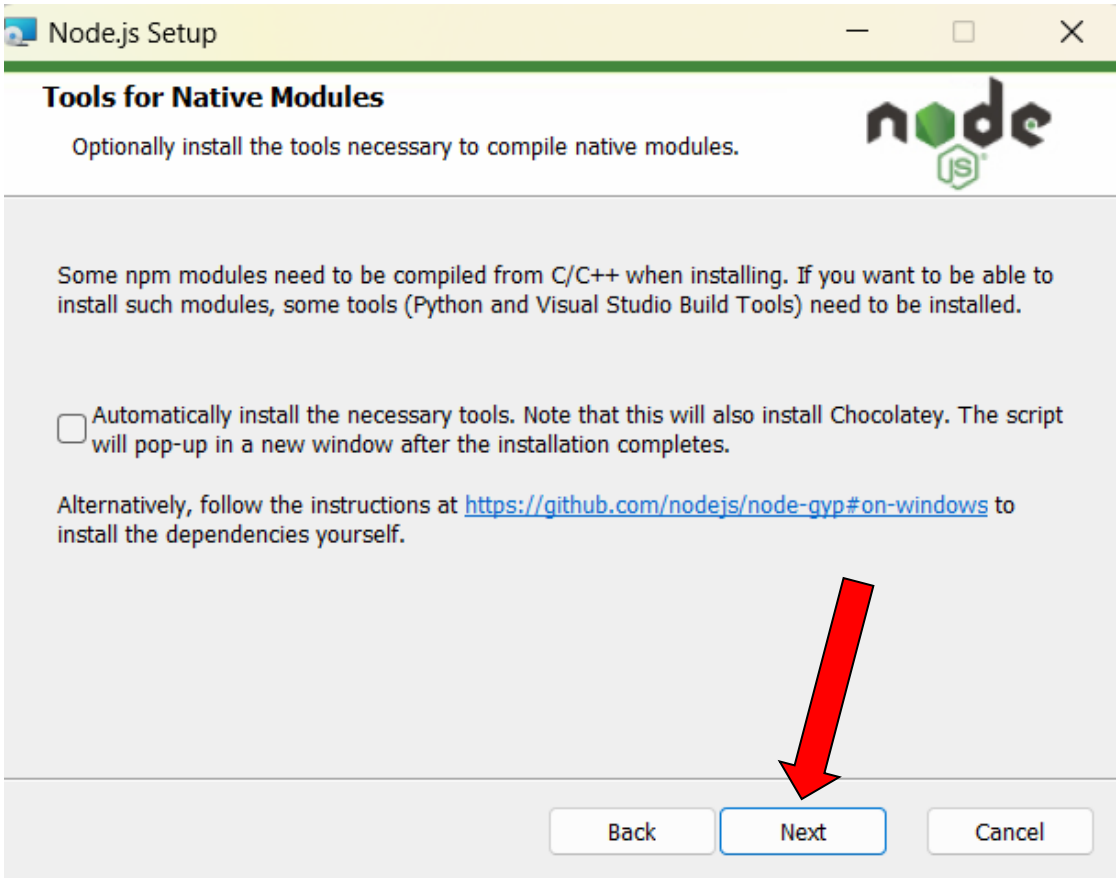


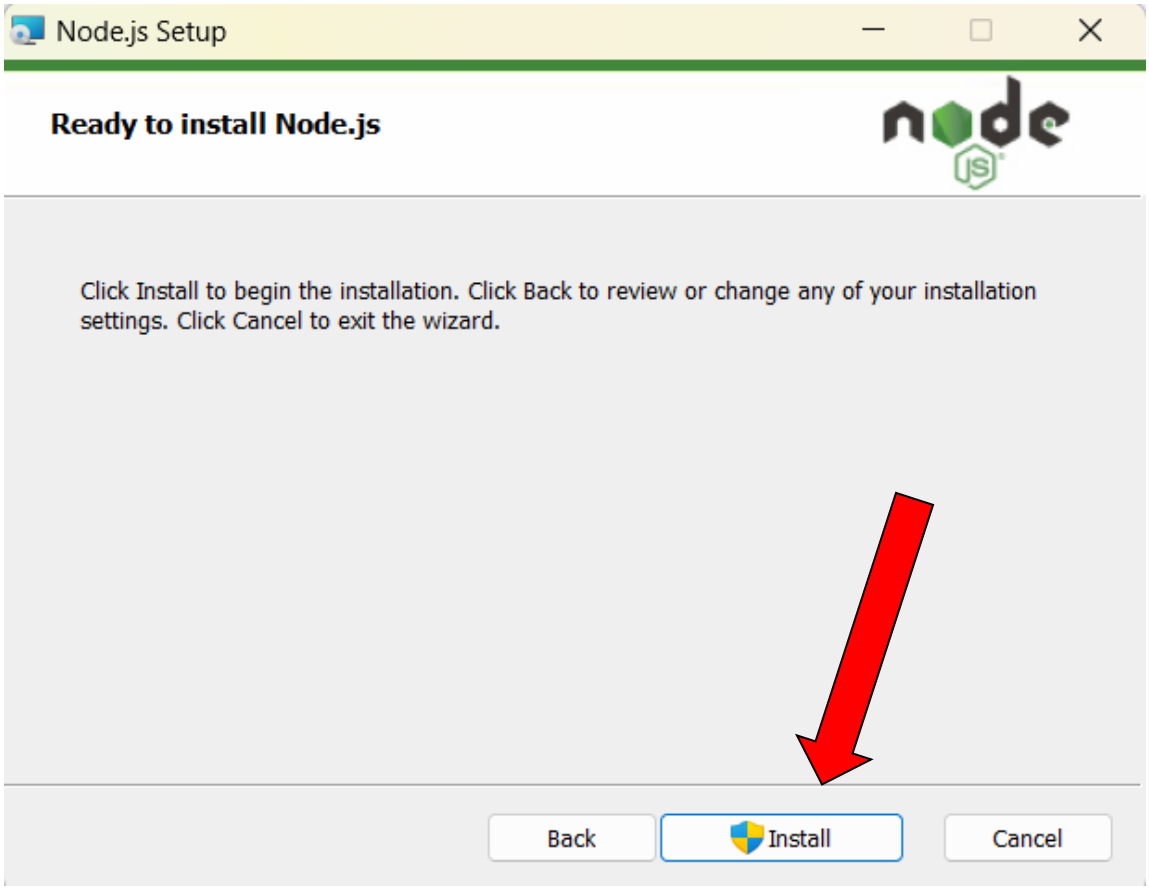
- Click on **I accept the Terms in the License agreements.**



- Choose Directory for installation but try to install in the default directory and proceed with **Next** Button.







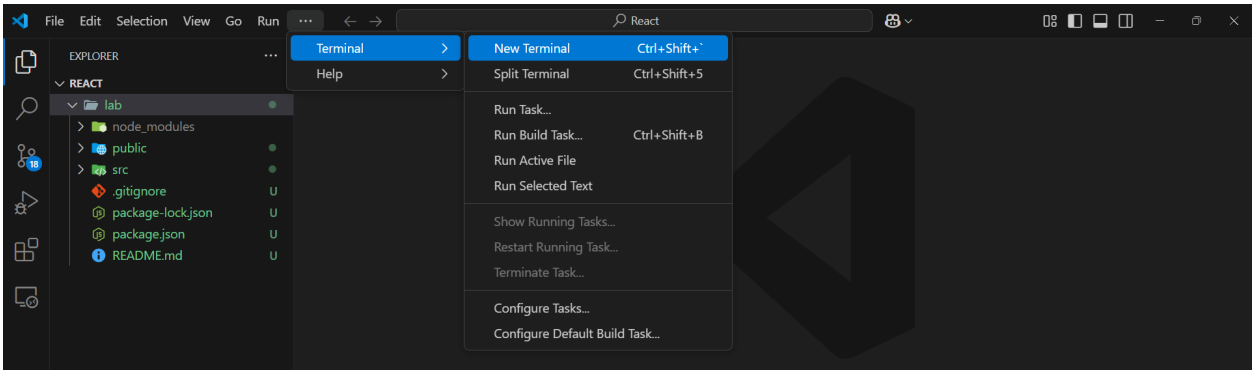
- A pop window will appear to allow it to make changes.
- Then Click on **Finish.**

Now, Open Command Prompt and run these commands to verify installation.

node -v

After Installing **Node.js** Create Folder and open it with **VS Code.**

- Open New Terminal and terminal will appear at bottom.



- Change terminal to cmd if powershell is running.

Now, in the terminal run the following command.

npx create-react-app projectname // Projectname can be of your choice.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   node

Microsoft Windows [Version 10.0.22631.4602]
(c) Microsoft Corporation. All rights reserved.

E:\Fall 2024\Old Folders\Web Engineering\Labs\Solutions\React>npx create-react-app lab10
```

It will take some time in installation and automatically generates a well-structured project folder with all necessary files.

Now, move to folder directory you created using command

cd projectname // Replace projectname with name of app you created.

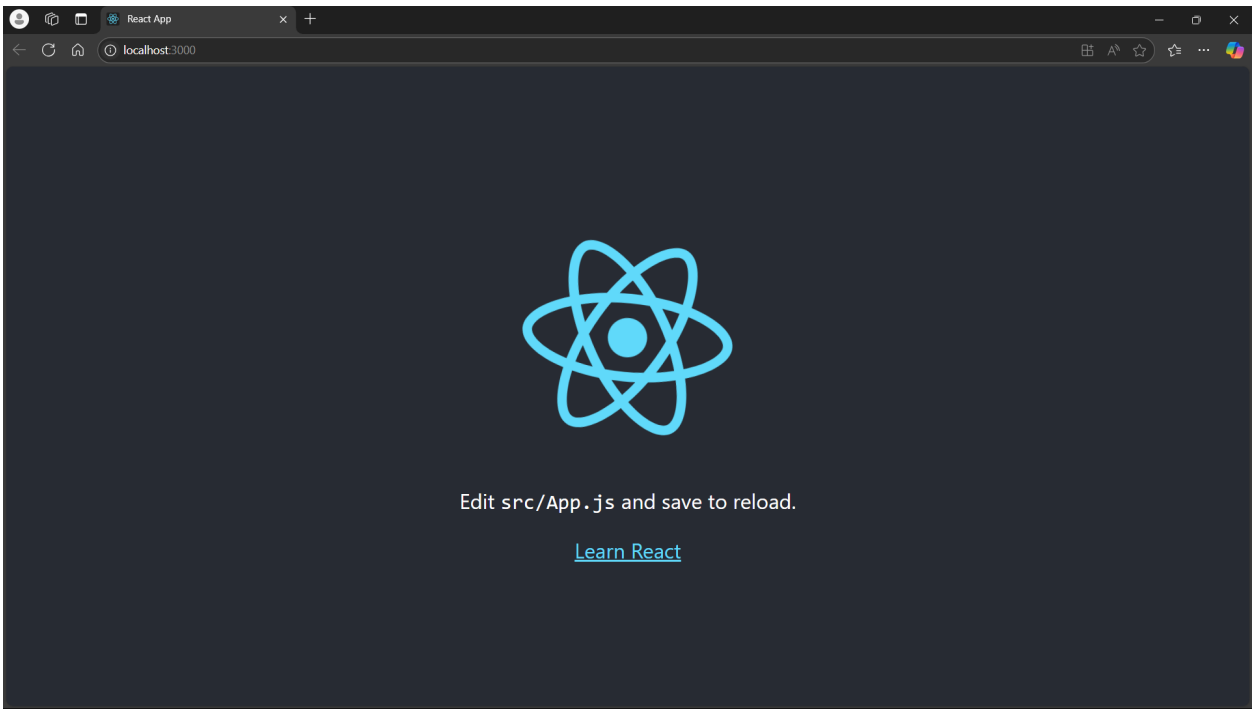
```
E:\Fall 2024\Old Folders\Web Engineering\Labs\Solutions\React>cd lab10
```

Now, run the app using **npm start** as it will start the development server.

```
E:\Fall 2024\Old Folders\Web Engineering\Labs\Solutions\React\lab10>npm start
```

Server will start after compilation and open the app in the browser on the local host.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   powershell
                                                                                       node

Compiled successfully!

You can now view lab11 in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://192.168.56.1:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```



4. Configure Git

Requirements

- Visual Studio Code
- Git (download and install from their official website)
- GitHub Account
- Node.js installed

Open terminal in VS Code and set your username and email:

```
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
```

Create Repository on GitHub

- Open GitHub
- Click on New Repository
- Enter repository name (e.g., react-app)
- Create repository without initializing it

Create React Application

- In VS Code terminal, run (ignore if already done):

```
npx create-react-app my-app
cd my-app
```

- Run the app to verify (ignore if already done):

```
npx create-react-app my-app
cd my-app
```

- Run the app to verify (ignore if already done):

```
npm start
```

- Initialize Git in Project

```
git init
git add .
git commit -m "Initial commit"
```

- Link Local Project to GitHub (Copy repository URL from GitHub and run)

```
git remote add origin https://github.com/username/react-app.git
git branch -M main
```

- Push Project to GitHub & verify that all project files are uploaded.

```
git push -u origin main
```

5. Tailwind CSS Installation (React)

- Install Tailwind and Dependencies

```
npm install -D tailwindcss@3.4.0 postcss autoprefixer
npx tailwindcss init -p
```

- Configure Tailwind (Open tailwind.config.js in root and update)

```
/** @type {import('tailwindcss').Config} */
export default {
  content: [
    "./src/**/*.{js,jsx,ts,tsx}",
  ],
```

- Add code to (postcss.config.js)

```
export default {
  plugins: {
    "@tailwindcss/postcss": {},
    autoprefixer: {},
  },
```

- Add Tailwind to CSS (Open src/index.css and add)

```
@tailwind base;
@tailwind components;
@tailwind utilities;
```

- Run the app and Add this in App.js to verify:

```
<h1 className="text-3xl font-bold text-blue-500">
 Hello, Tailwind Works
</h1>
```

Lab Task

1. Install Node.js and verify the installation.
2. Create a new React application.
3. Explore the folder structure of the React project.
4. Run and test the application.
5. Install and configure Tailwind CSS
6. Configure and push files to your GitHub repository

Submit only Pdf File with screenshot of Installation, verification & output after running your app along with GitHub URL + Screenshot. Output must show usage of Tailwind in your App.

Lab Assessment Criteria

Lab # 10 Marks Distribution

	ER1	ER6	RR1
Task	6 Points	2 Points	2 Points

Lab # 10 Rubrics Evaluation Guidelines

#	Qualities & Criteria	0 < Poor <=4	4.1 < Satisfactory <= 7	7.1 < Excellent <=10
ER1	**Task Completion**	Minimal or no program functionality was achieved**.**	Some tasks were completed, but the program had errors or was incomplete.	All tasks were completed, and the program runs without errors.
#	**Qualities & Criteria**	**0 < Poor <=4**	**4.1 < Satisfactory <= 7**	**7.1 < Excellent <=10**
ER6	**Program Output**	Output is inaccurate or poorly presented.	Output is mostly accurate but may lack labels, captions, or formatting.	Output is clear, accurate, and well presented with labels, captions, and proper formatting.
#	**Qualities & Criteria**	**0 < Poor <=4**	**4.1 < Satisfactory <= 7**	**7.1 < Excellent <=10**
RR1	**Report Structure and Clarity**	Report is poorly organized, difficult to navigate, and lacks professional formatting.	Report is mostly organized but may have some minor structural issues or formatting inconsistencies.	Report is well-organized with a clear logical flow, easy navigation, and professional formatting.